

problem4

May 6, 2022

0.1 Problem 4 - Deep Reinforcement Learning

This question is based on Deep RL concepts discussed in Lecture 8. You need to refer to the papers by Mnih et al., Nair et al., and Horgan et al. to answer this question. All papers are linked below.

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[1]: from IPython.display import Markdown, display  
  
display(Markdown("./problem4/4_1.md"))
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1 4

1.1 1

Episodic And Non-Episodic Environment

Episodic: Non-Sequential Environment, current action will not affect the next action. In RL, episodes are considered agent-environment interactions from initial to final states. In a car racing video game, you start the game (initial state) and play the game until it is over (final state). This is called an episode

Continuous: Current action will affect the next action. When we play the poker game, current action will affect the next action.

1.2 2

What do the terms exploration and exploitation mean in RL ? Why do the actors employ ϵ -greedy policy for selecting actions at each step? Should ϵ remain fixed or follow a schedule during Deep RL training ? How does the value of ϵ help balance exploration and exploitation during training.

Exploration: explore the action space and get the reward corresponding to the action.

Exploitation: exploit the action space and choose the action with the highest Q value.

Epsilon-greedy policy: We takes action using the greedy policy with a probability of $1 - \epsilon$ and a random action with a probability of ϵ . This approach ensures all the action space is explored.

ϵ should follow a schedule during training, but for burn-in period, we may use a fixed value.

Larger ϵ means more exploration, but less exploitation.

1.3 3

How is the Deep Q-Learning algorithm different from Q-learning ? You will follow the steps of Deep Q-Learning algorithm in Mnih et al. (2013) page 5, and explain each step in your own words. (2)

The biggest difference between DQN and Q learning is the Q table. In Q learning, this is a table. Enter (s, a) to query the corresponding Q value. In DQN, this is a function that is replaced by a neural network. Input (s, a) can output the corresponding Q value.

Update Steps:

- $Q(s, a|\theta)$ is the Q function.
- For current state, we have $Q' = Q(s, a|\theta)$. We can use MC or TD to get the Q' as the target Q.
- Then we use $Q' - Q$ as the loss to update the Q function.

1.4 4

What is the benefit of having a target Q-network?

Neural Network can be seen as a **compressed** version of the Q table, especially for infinite state space. Thus we can use the target Q-network to estimate the Q-value of the given state-action pair, which can be used to get the best action given the current state.

1.5 5

How does experience replay help in efficient Q-learning?

This means instead of running Q-learning on (state, action) pairs as they occur during simulation or actual experience, the system stores the data discovered as (state, action, reward, next_state).

The experience replay uses the stored data to train the Q-network, which means the past experience can be sampled multiple times to train the Q-network.

1.6 6

What is prioritized experience replay and how is priority of a sample calculated?

Prioritized Experience Replay is a type of experience replay with more frequently replay transitions with high expected learning progress, as measured by the magnitude of their temporal-difference (TD) error.

For the sampling transition i , $P(i) = \frac{p_i^\alpha}{\sum_k p_k^\alpha}$, α controls the priority of the transition.

1.7 7

Compare and contrast GORILA (General Reinforcement Learning Architecture) and Ape-X architecture. Provide three similarities and three differences. (3)

Similarity - Both of GORILA and Ape-X architecture have two parts: acting and learning. - Both have a replay memory. - Both have a parameter server.

Differences - GORILA's Replay Buffer is off-line, Ape-X's Replay Buffer is on-line.(Replay Experiences) - The replay memory is centralized in Ape-X, while GORILA can be distributed. - Ape-X is not uniformly sampling (Ape-X use prioritized experience replay), but GORILA is uniformly sampling.

1.8 8

Why the performance improves with number of actors in Ape-X?

Actor is used to sample and generate experiences from the environment, more actors means more exhaustive sampling.